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ITECH3102 – Data Visualization
Week 6

Lab Exercise – Complex data visualizations (1)

ID: _____Name: _____

In previous projects, we discussed how to visualize simple data in order to answer some questions. However, some applicational cases may involve more complex questions that cannot be easily answered by **directly using existing fields.** In this project, we learn basic approaches of data visualization and discuss how to visualize complex data. Here visualizing complex data indicates that visualization cannot be easily achieved through **directly using existing fields** from raw data sources because these fields don't exist in raw data sources. Instead, for visualizing complex data, we may require the use of some more-advanced techniques. For example, we may need to take simple computation on the existing fields or to combine the existing fields or to use other methods in order to make new dimensions or measures that don't exist in raw data source.

In particular, we discuss some main approaches to visualize complex data and enhance data: changing the type of aggregation, creating new calculated fields, or setting up parameter controls.
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Project 1: Using aggregate functions

Tableau provides users with some functions. For example, the types of aggregation supported by Tableau include the following:

Sum

This aggregation adds up the measure's values and returns the sum of all the values in the expression. SUM can be used with numeric fields only. NULL values are ignored.

Average

This aggregation gives you the arithmetic mean of the measure's values.

The AVG function returns the average value for an expression. It is calculated as the sum of all of the numbers for the expression divided by the count of the number of records in that expression. For example, if you have a set containing (24, 30, 15, 5, 18) the average of those five numbers will be: (24 +30+15+5+16)/5 =15.

Median

This aggregation gives you the median of the measure's values.

Count and Count (Distinct)

These two aggregations count the occurrences of different values in two ways. Imagine that 1000 data rows underlie a specific mark in your view. Each row records a single sales transaction, and one field contains the type of product transacted (out of 30 different types of products that came up in these 1000 transactions). Using Count on the field Product Name will return the number of products transacted: 1000. Using Count (Distinct), on the other hand, returns the value 30; only the unique occurrences are counted. When you use a live connection to MS Access, Count (Distinct) is unavailable; you must create an extract first.

Minimum and Maximum

These aggregations return the smallest and the largest value, respectively, of a measure or a dimension.

MIN

The MIN function is very similar to the MAX function, whereby this function returns the minimum value when applied to a single field in an aggregate calculation. The MIN

function can also be applied to return the minimum of two arguments. These arguments must be of the same type.

MAX

The MAX function is normally reserved for numbers; however, this function can also be used on strings within Tableau. When MAX is applied to strings, the MAX value returns the string that is highest within the data's sort sequence for that particular string.

Percentile

This aggregation returns the value for the selected percentile. Percentile is not available for a live connection, so you first need to create an extract.

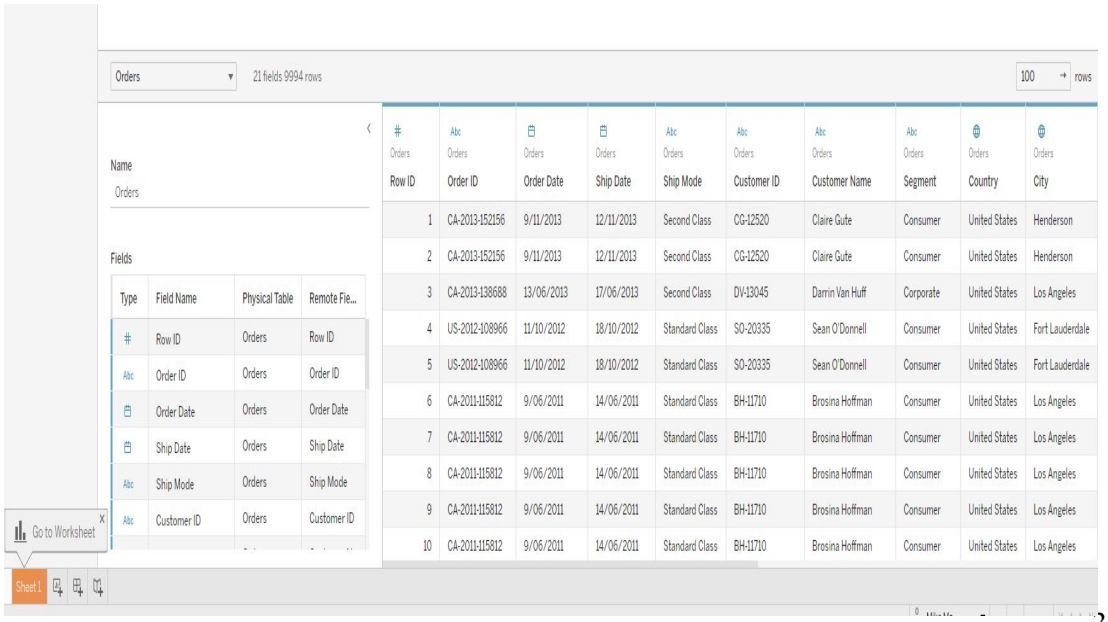
Variance and Variance (Pop.)
These aggregate functions return the sample variance and population variance, respectively.

Download Dataset: order-information.xls from week 6 lab in Moodle

Step 1. Load the data into Tableau and check that your data has been loaded correctly
Step 2 Create a bar chart

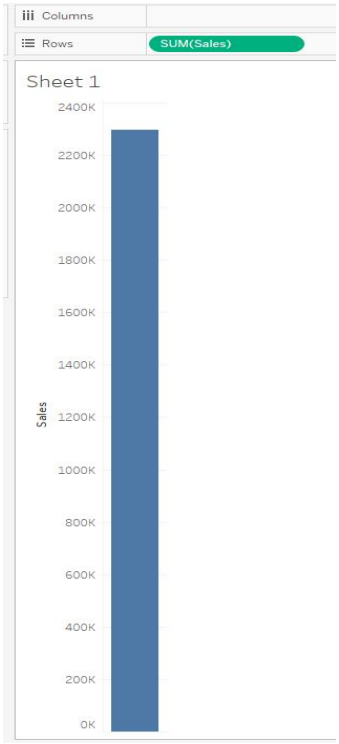
Step 1. **Load the data into Tableau**

Drag Order field from the Measures section in the data pane on to center of canvas and drop it.



After clicking Sheet 1 tap on the bottom, the data will be connected to your computer and shown in the data pane on.

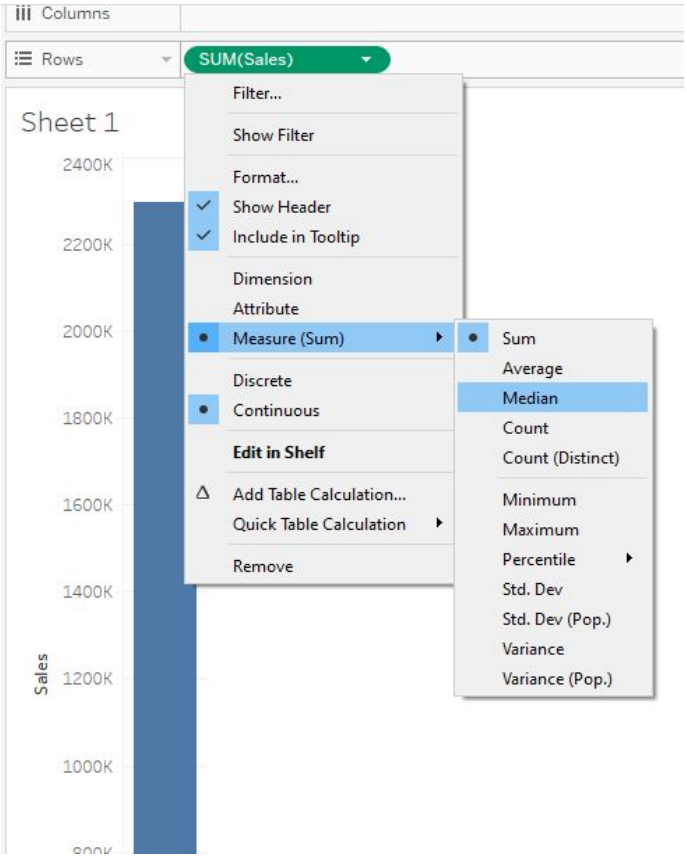
Drag and drop **Sales** field from the Measures section in the data pane on to the Rows shelf. You now have a bar chart with a single bar representing the sum of sales (for all of the data in the data source).



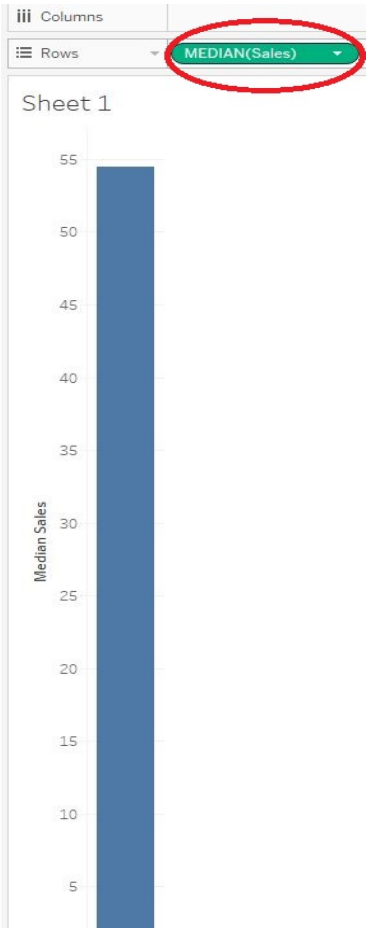
By default, Tableau uses Sum as the aggregation. But this is not always the most appropriate way to aggregate the data. So, we may need to use a different function.

Step 1. **Use a different function**

Now we click the small arrow on the SUM(Sales), we get



After selecting Median, we get



Project 2: Using calculated fields

When visualizing your data, you are not restricted to the dimensions and measures in your data source. There are times when the fields available in your data won't be enough to fulfill your requirements. For example, your data may have fields such as population and area. However, you may want to find out how much population density in an area is. In this case, you can add your own calculated fields. These calculations are saved as new fields in the data model of your workbook and can be used as dimensions or measures in your visualizations.

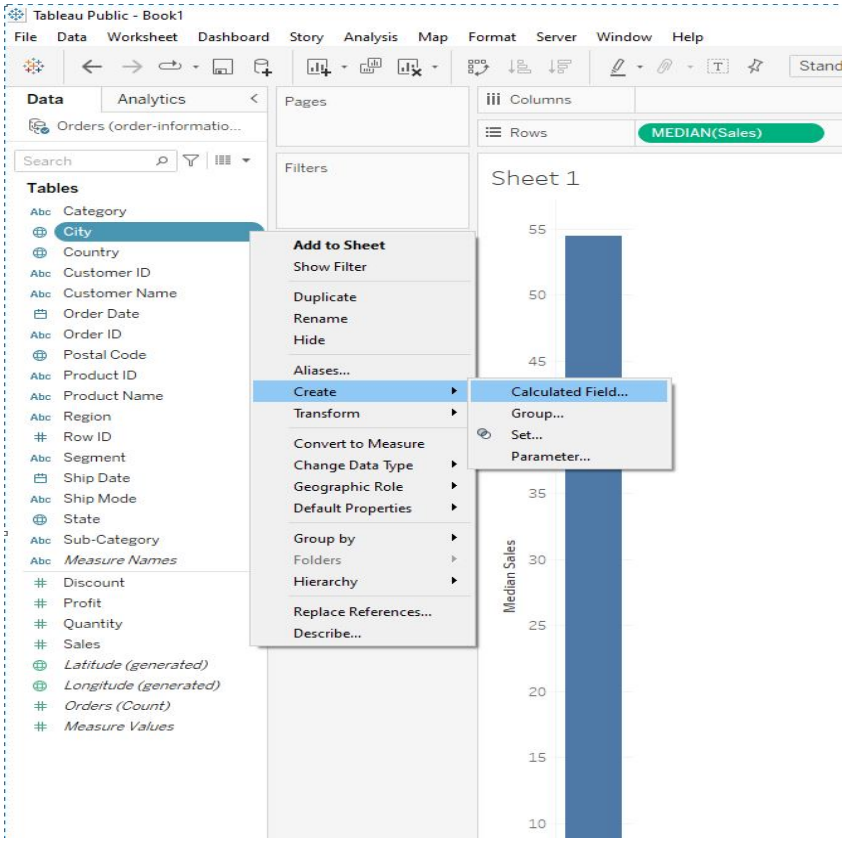
We can classify our calculations into categories as follows:

- String Calculations
- Arithmetic Calculations
- Date Calculations
- Logic Calculations

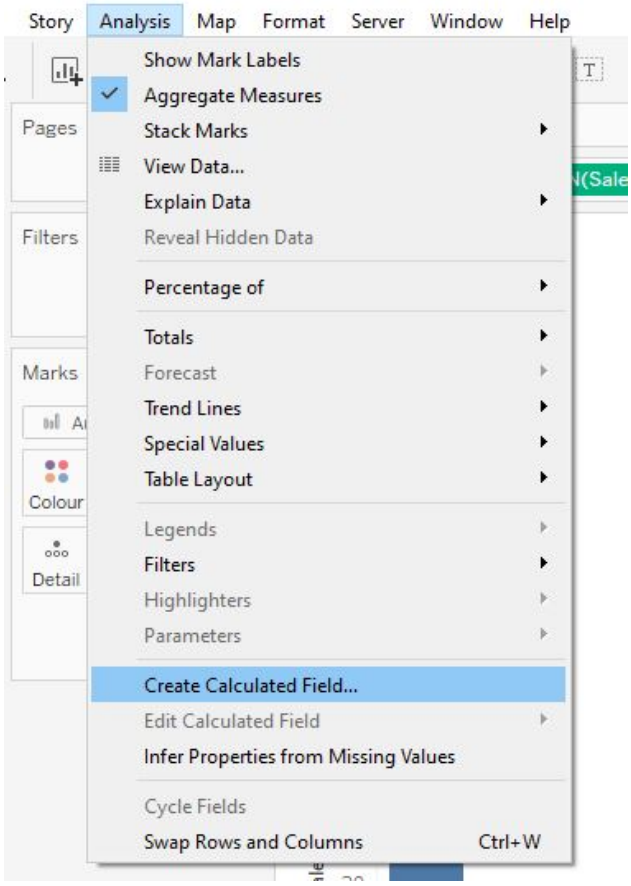
In this project, we focus on arithmetic calculations and logic calculations.

1. Create a calculated field

Now let us to create a calculated field. We can click the small arrow at the top of the Data pane or right-click on any Dimension or Measure and select the option of Create | Calculated Field.



We can also create a calculated field by selecting Analysis | Create Calculated Field:



Once clicking Create Calculated Field, we will get a new dialog box where we give a name to our calculated field and enter codes or formulae in code editor area.



We are now showing examples of using calculated fields.

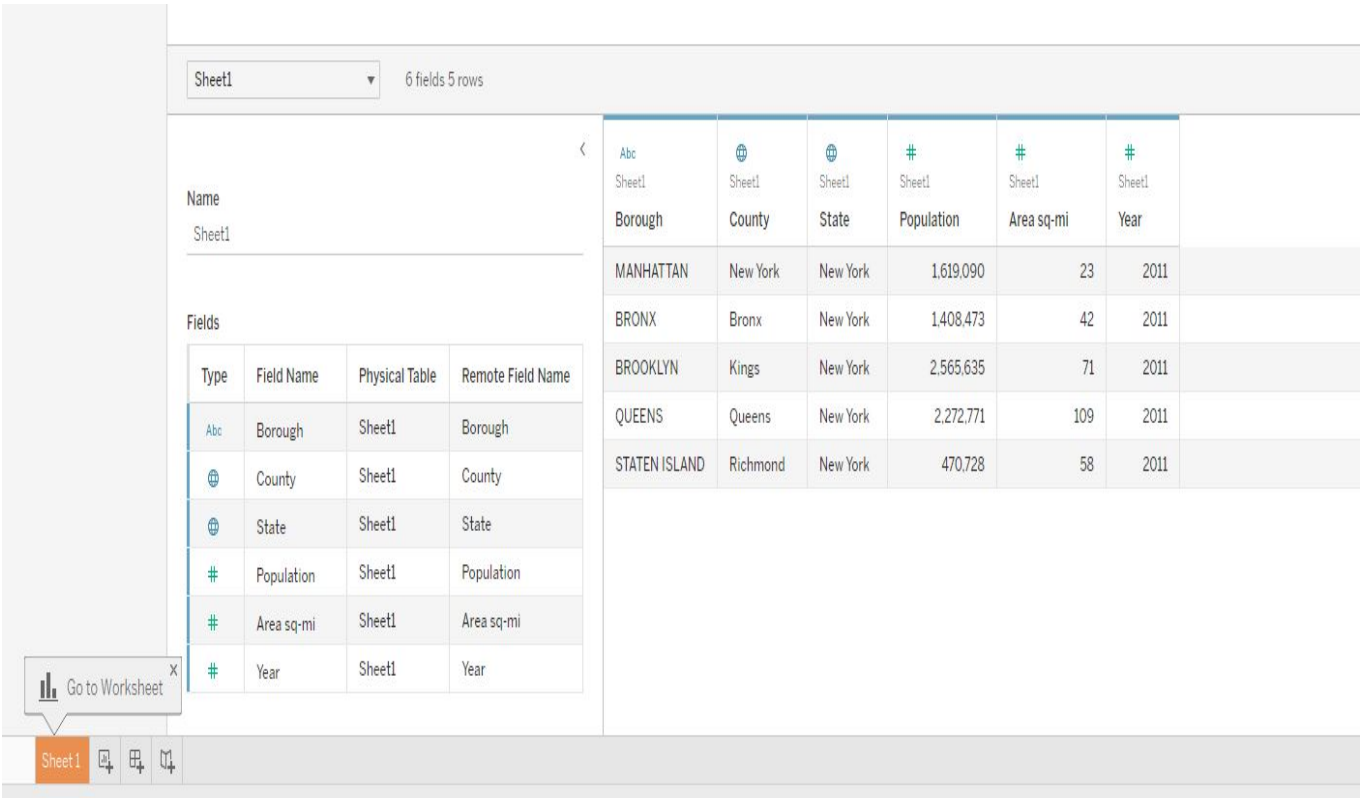
2. Create an arithmetic calculation

Task: The given dataset provides information about population and areas in a city. We want to find out how much population density in an area is from this data. However, population density is not available from existing the dimensions and measures in this data source. The goal of this task is to create calculated field for displaying population density of a district.

Download Dataset: simple-data-bogoughs.xlsx from week 6 lab in Moodle

Step 1. Load the data into Tableau and check that your data has been loaded correctly.
Step 2. Create a calculated field.
Step 3. Create a filled map to display population density.
Step 4 Edit location
Step 5 Add label to map

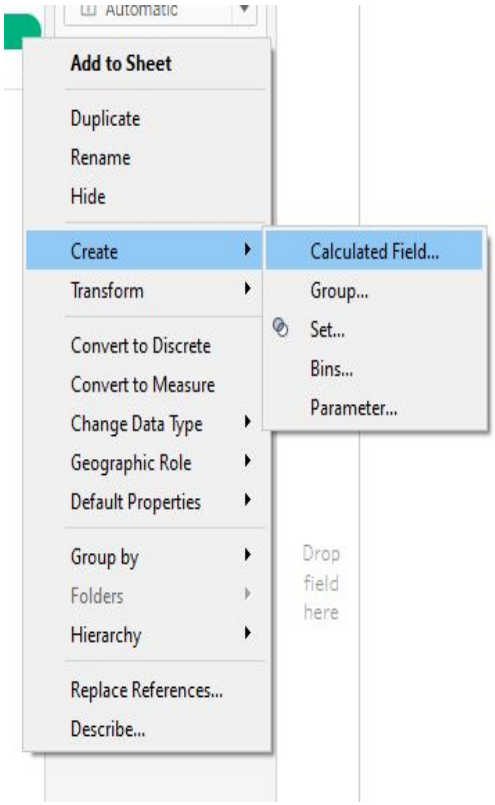
Step 1. Load the data into Tableau and check that your data has been loaded correctly.



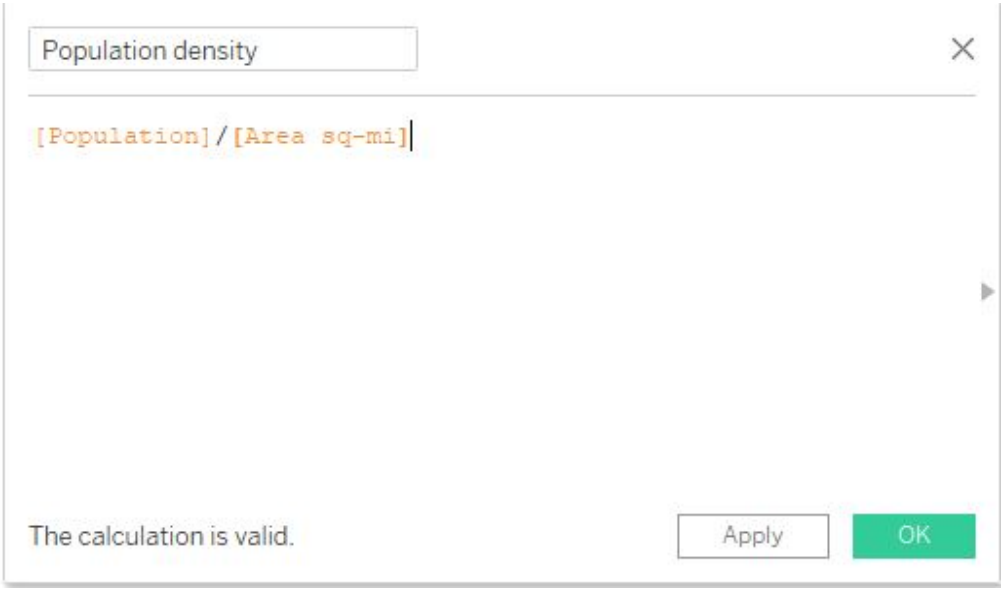
After clicking Sheet 1 tap on the bottom, the data will be connected to your computer and shown in the data pane on.

Step 2. Create a calculated field.

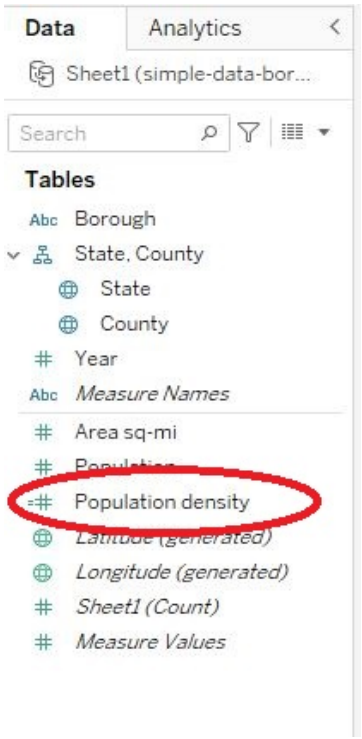
Population density can be expressed as formula of population being divided by area.
Right click any location in Dimensions or Measures area in Data pane and select Create Calculated Field.



Then enter the name of the calculated field and enter code in code editor area to perform the density calculation as following.

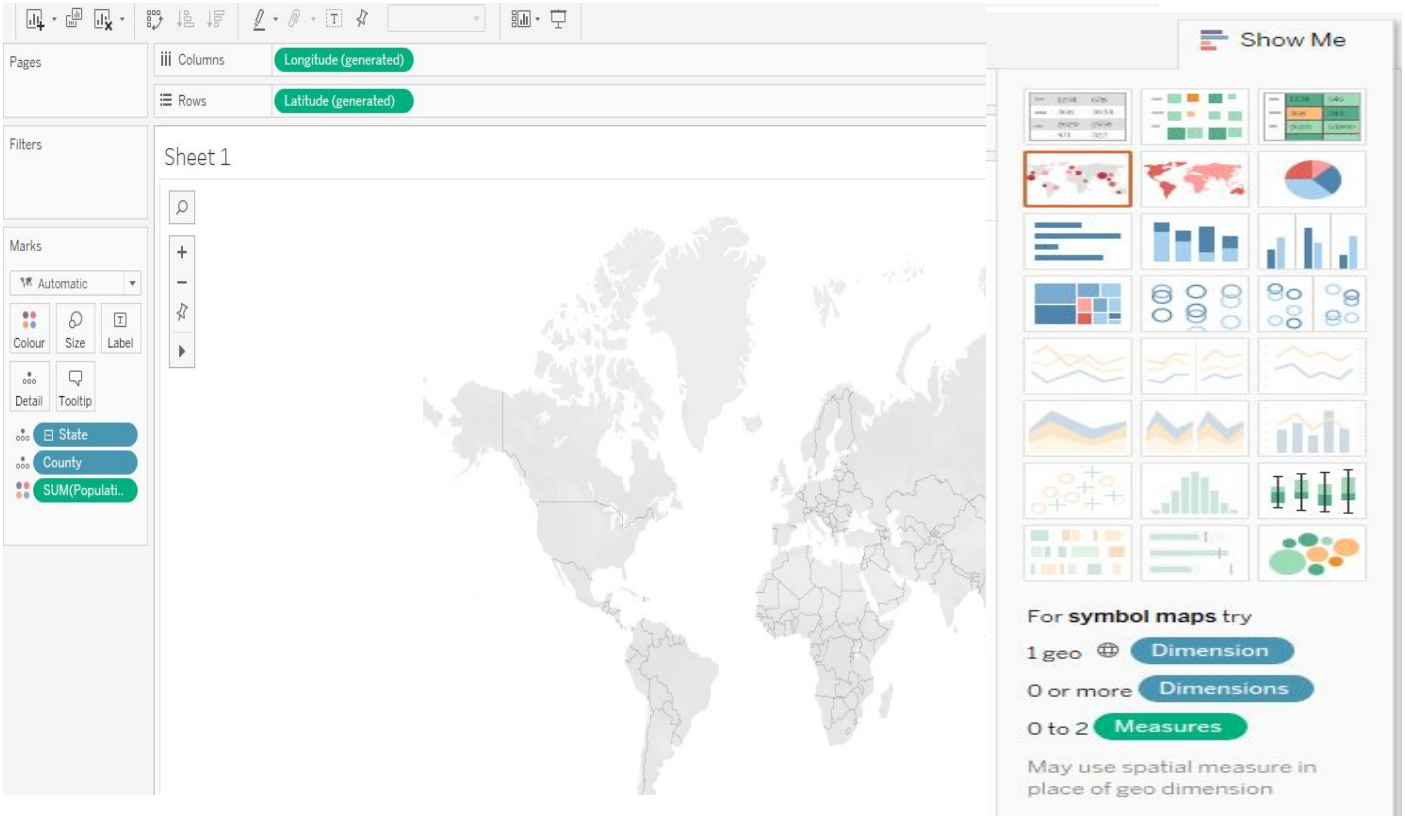


After click OK button, a new field of Population density you just created appears in the Measures area.



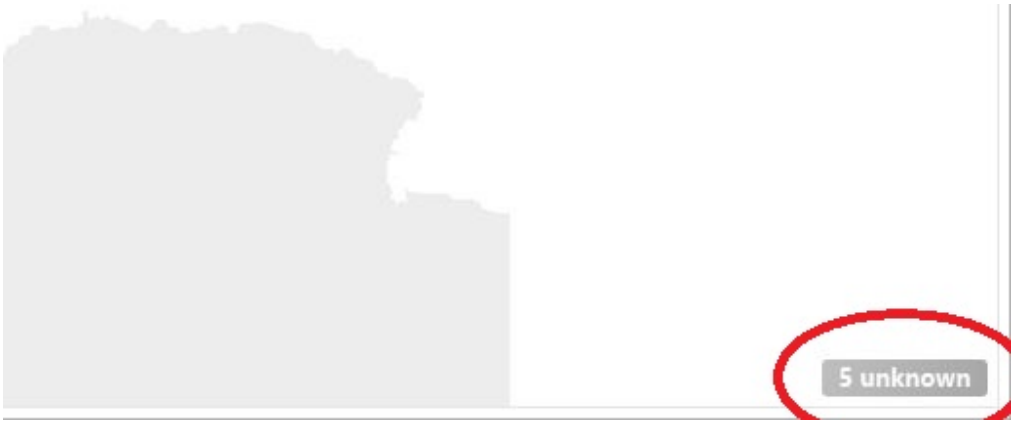
Step 3. Create a filled map to display population density.

Select County, hold down the Control key and then select Population Density and select filled map in Show Me card

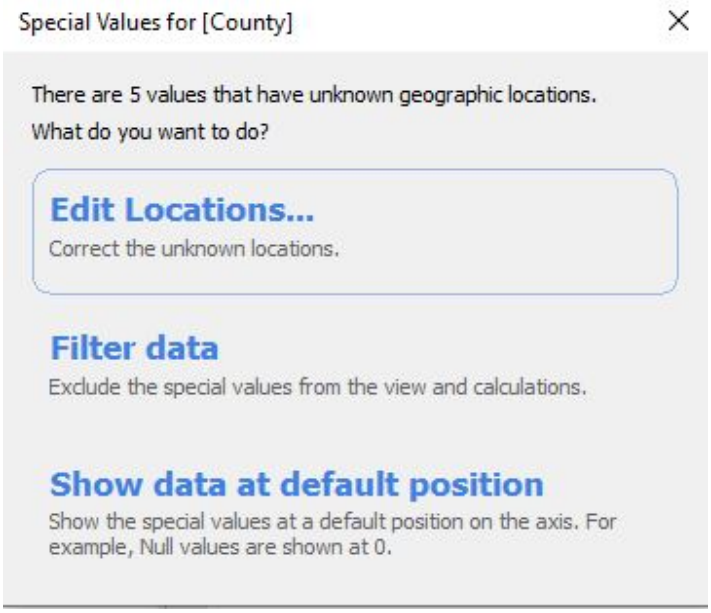


Step 4 Edit location

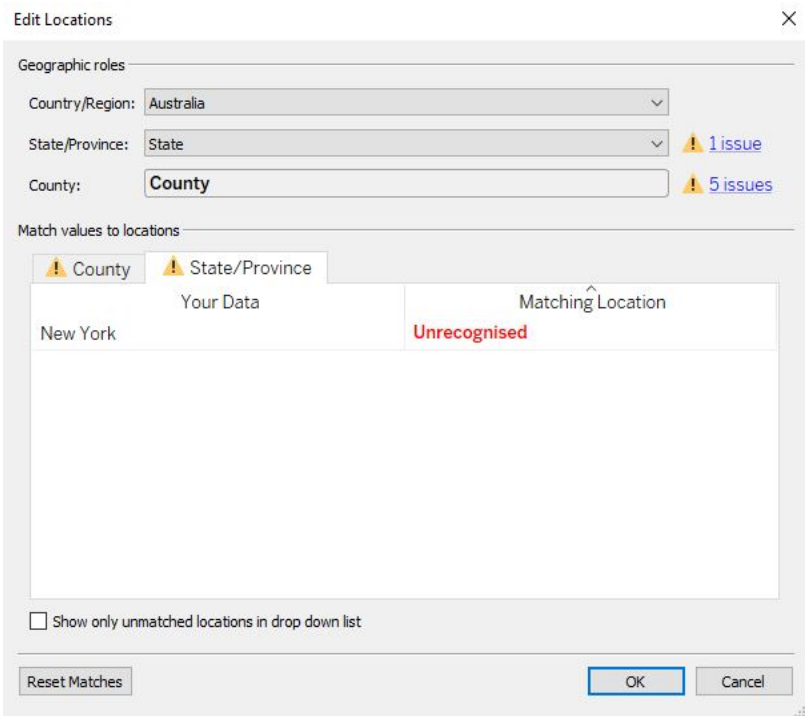
In the lower-right area of the map, there is a gray text (5 unknown). This indicates that five geographic records are unrecognized in the dataset. So, we need to change location from Australia to United States.



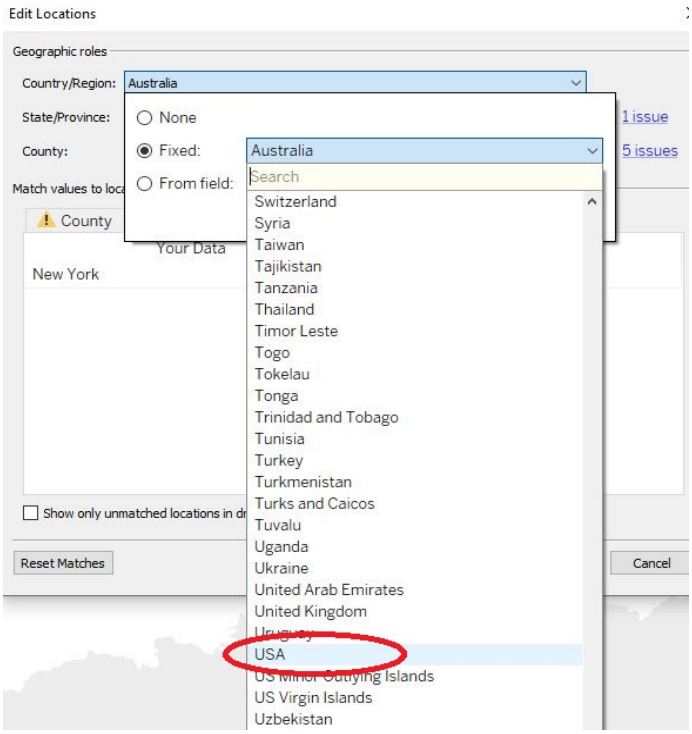
Click the gray text 5 unknown to open a menu

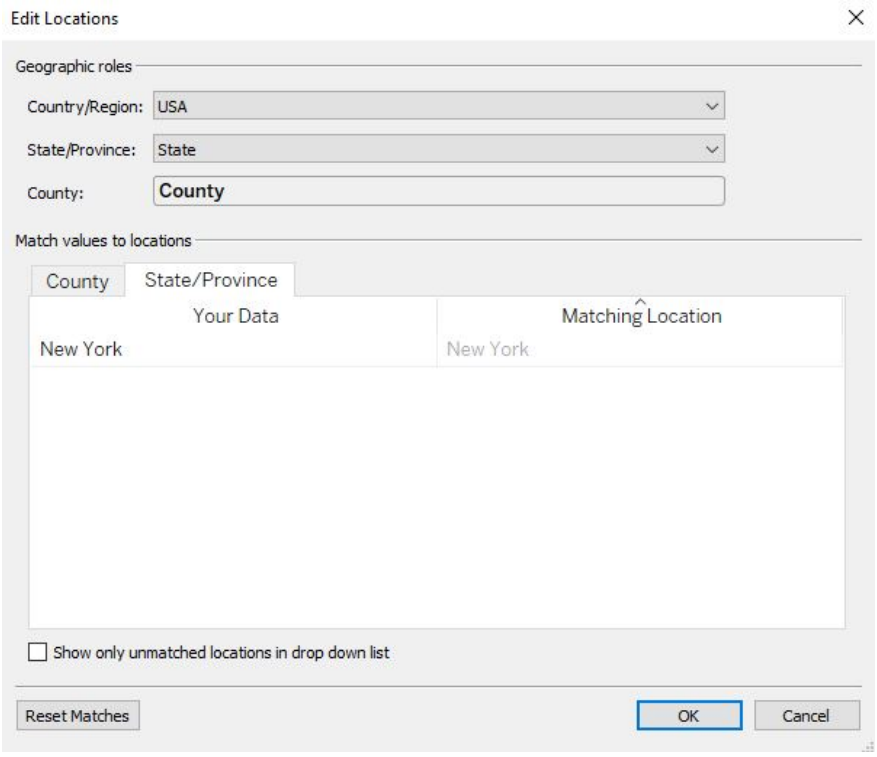


Click Edit Location to change location

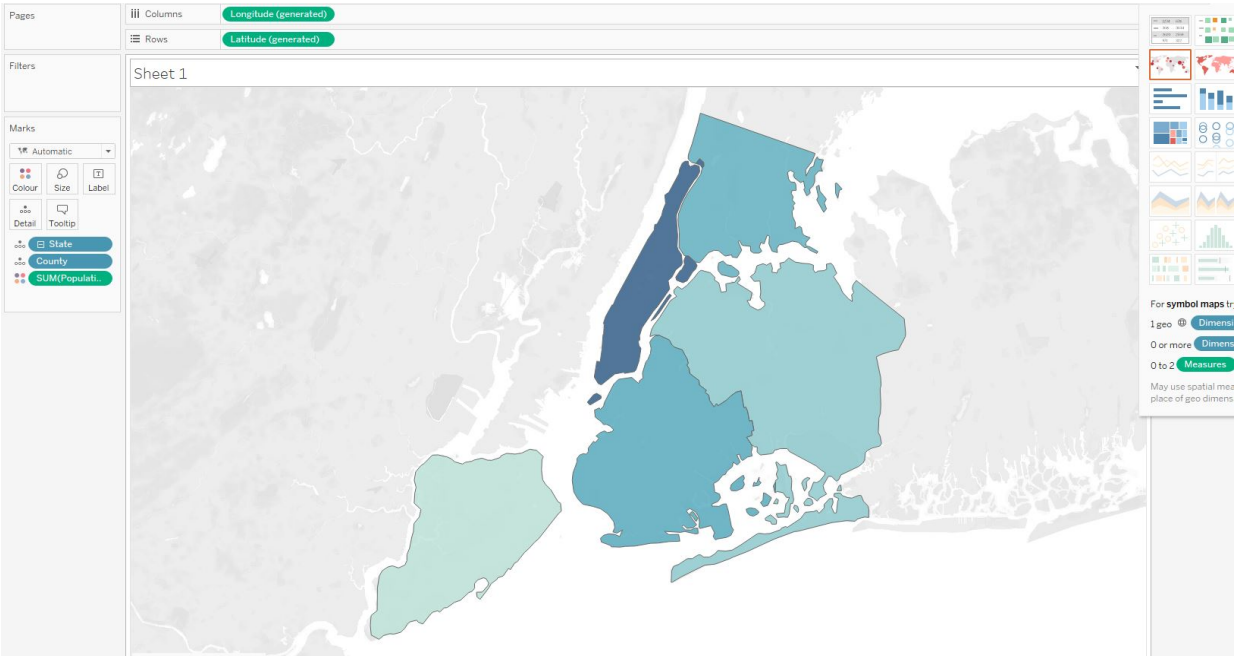


Location New York is highlighted as Unrecognized with red color, so we need to change Country/region from Australia to USA.



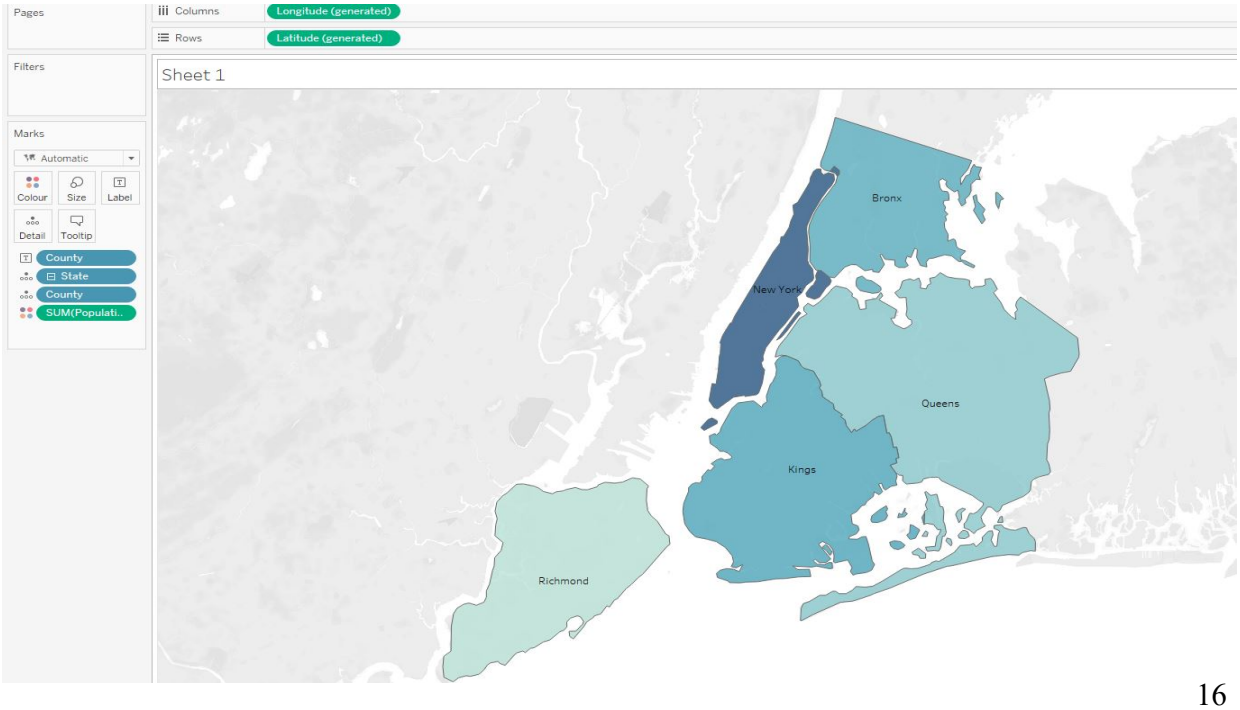


Then press Ok.



Step 5 Add label to map

Drag County from Data pane to Label in the Marks card.



Discussion

Based on the above filled map, which county has the highest population density? Why?

Project 3. Create a logic calculation

We introduce you to the use of application logic, including case statement and if-then statements that you may be familiar with from various programming languages.

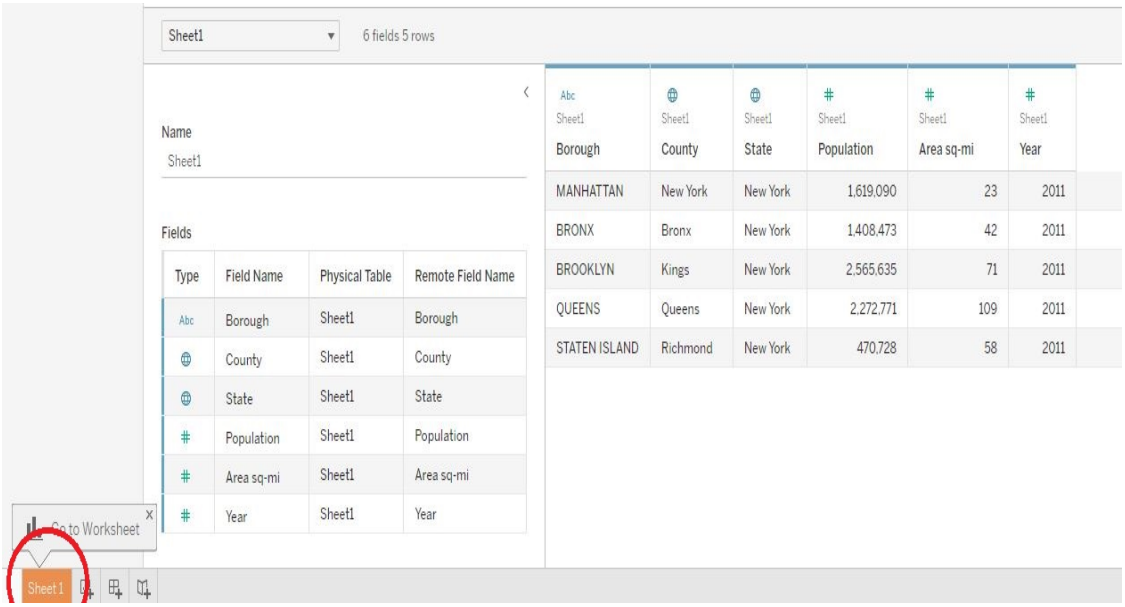
(1) Case Discrimination

The Tableau CASE and WHEN keyword is used to create a logical statement with dimension or measure variable(s) from the given data to see if it matches a certain condition(s).

CASE will return the value you specify as the return value in the statement.

Download Dataset: simple-data-bogoughs.xlsx from week 6 lab in Moodle

Step 1. **Load the data into Tableau** and check that your data has been loaded correctly



After clicking Sheet 1 tap on the bottom, the data will be connected to your computer and shown in the data pane on.

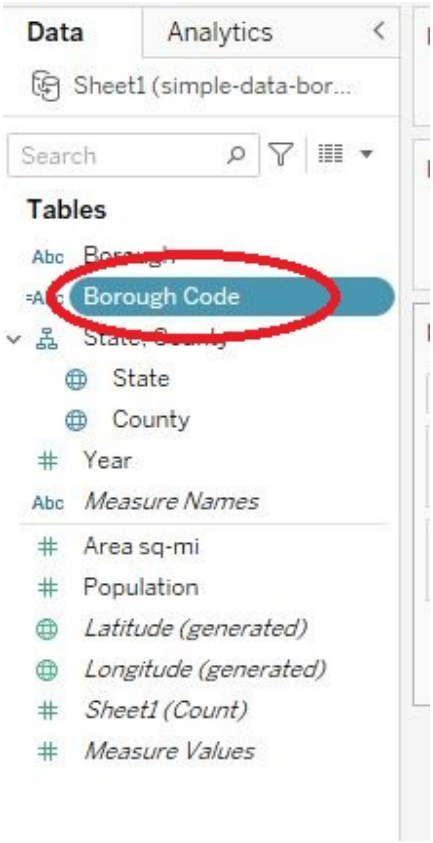
Step 2. Create a calculated field.

Right click any location in Dimensions or Measures area in Data pane and select Create Calculated Field.

Then enter the name of the calculated field and enter code in code editor area to perform a logic calculation as following.



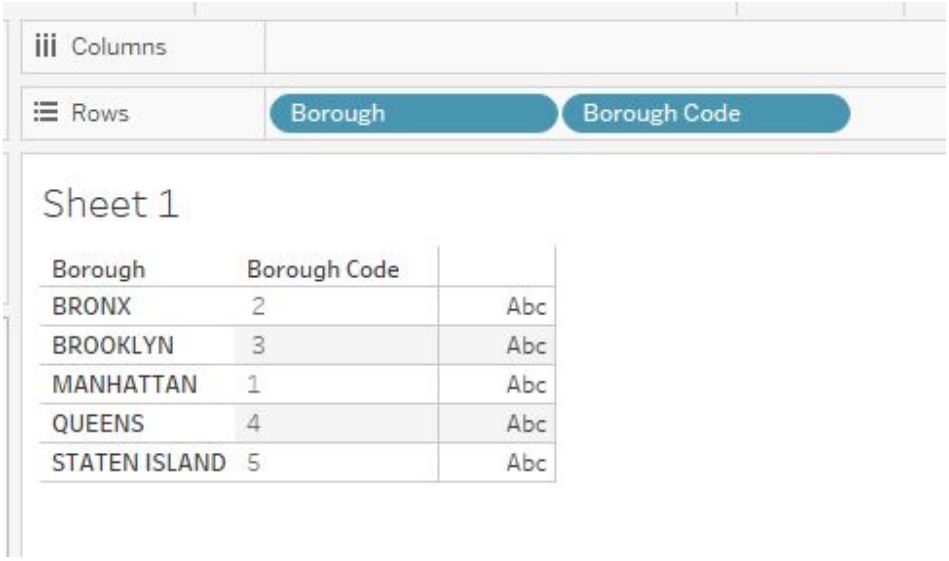
After clicking OK button, a new field of Borough Code you just created appears in the Dimensions area.



In the above logic calculation, The CASE statement in the calculated field initiates the case discrimination, in this case based on the dimension Borough. The WHEN-THEN pairs define the new value to be returned for each case. For example, the output is 1 for each record where the Borough value is MANHATTAN. Should none of the stated cases apply, the ELSE statement, which is optional, ensures that the text “not defined” is displayed.

Step 3. Display the result

Now drag the Borough and the Borough Code variables from Data pane into the Rows shelf of your Sheet.



(2) IF-THEN-ELSE logic

IF-THEN-ELSE logic can be used in calculated fields. While many problems can be solved with the simple CASE statement we just looked at, using IF-THEN-ELSE lets you include Boolean operators and nested logic.

For instance, a manager responsible for the product subcategories Copiers, Paper, and Fasteners might be interested in comparing those numbers to those of the other subcategories. You can add a new dimension containing the newly grouped subcategories.

Task: Using different colors represents various sales in different sub-categories.

Download Dataset: order-information.xls from week 6 lab in Moodle

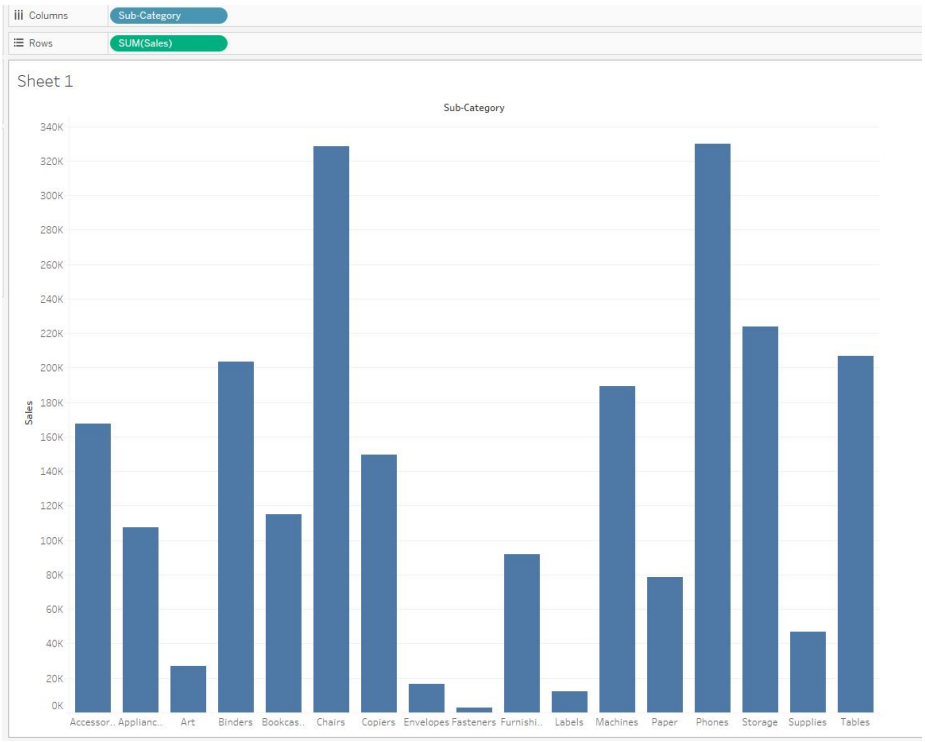
Step 1. **Load the data into Tableau**

Drag Order field from the Measures section in the data pane on to center of canvas and drop it.

After clicking Sheet 1 tap on the bottom, the data will be connected to your computer and shown in the data pane on.

Step 2. Create a bar chart.

Drag and drop the Sub-Category field from Data pane into the Columns shelf, followed by dragging and dropping the Sales field into the Rows shelf. This will create a vertical bar chart:



Y

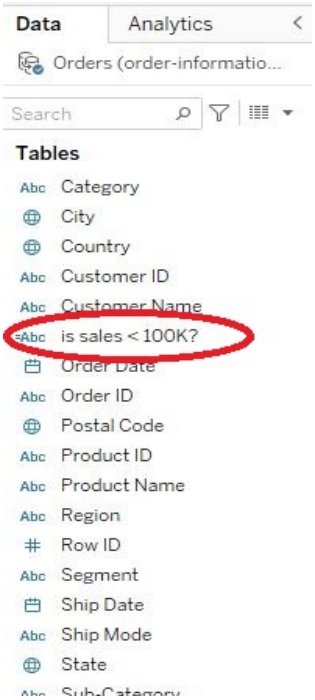
Step 3. Create a calculated field.

Right click any location in Dimensions or Measures area in Data pane and select Create Calculated Field.

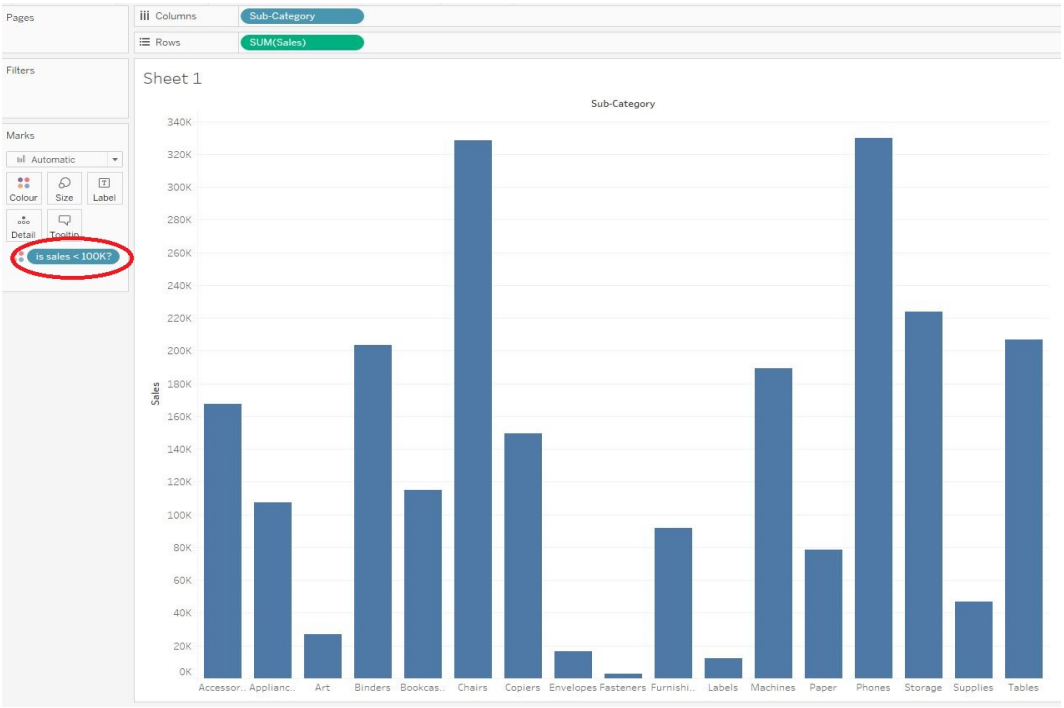
Then enter the name of the calculated field and enter code in code editor area to perform a logic calculation as following.



After clicking OK button, a new field of is sales < 100K? you just created appears in the Dimensions area.



Now drag calculated filed is sales<100K? from Data pane to Color shelf in the Marks card to color the bar:

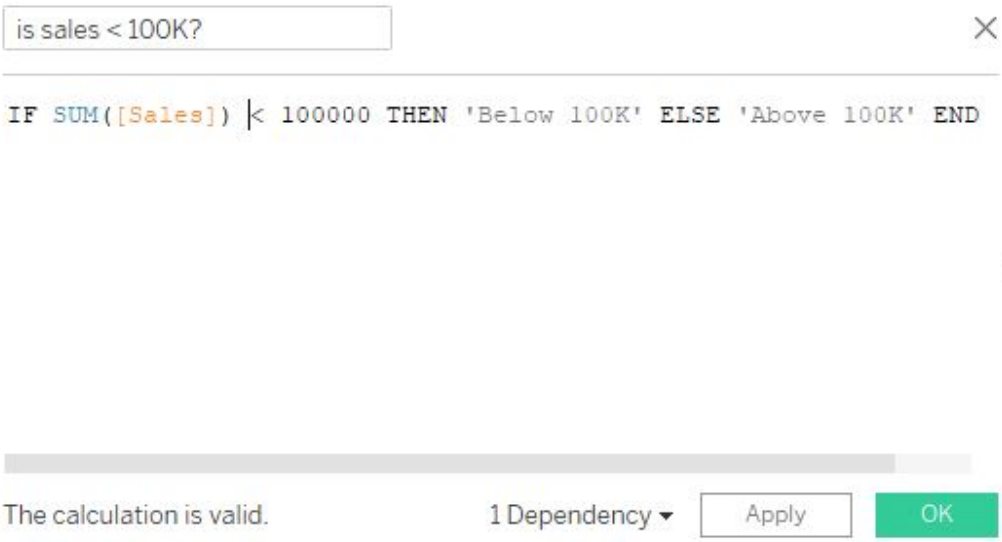


But the bar chart is the same as it was earlier because the colors on the graph have not changed. One of objectives in this task is that using different colors represents various sales in different sub-categories.

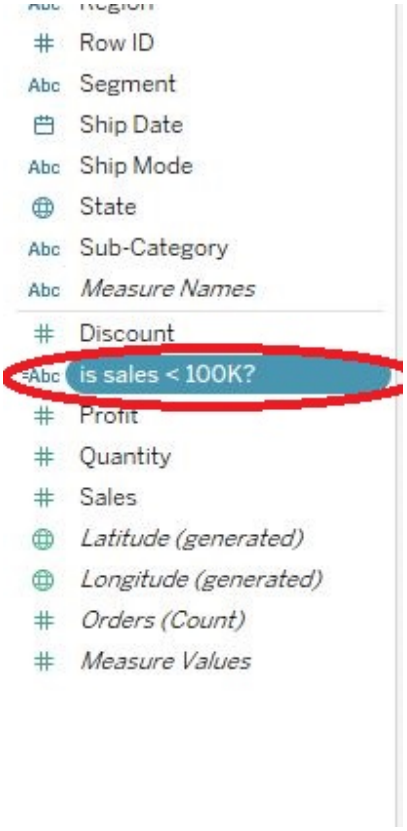
Step 4. Modify the calculated field

Click calculated filed is sales<100K? and select Edit to open code editor.

Then enter code SUM([Sales]) to replace [Sales] in code editor:

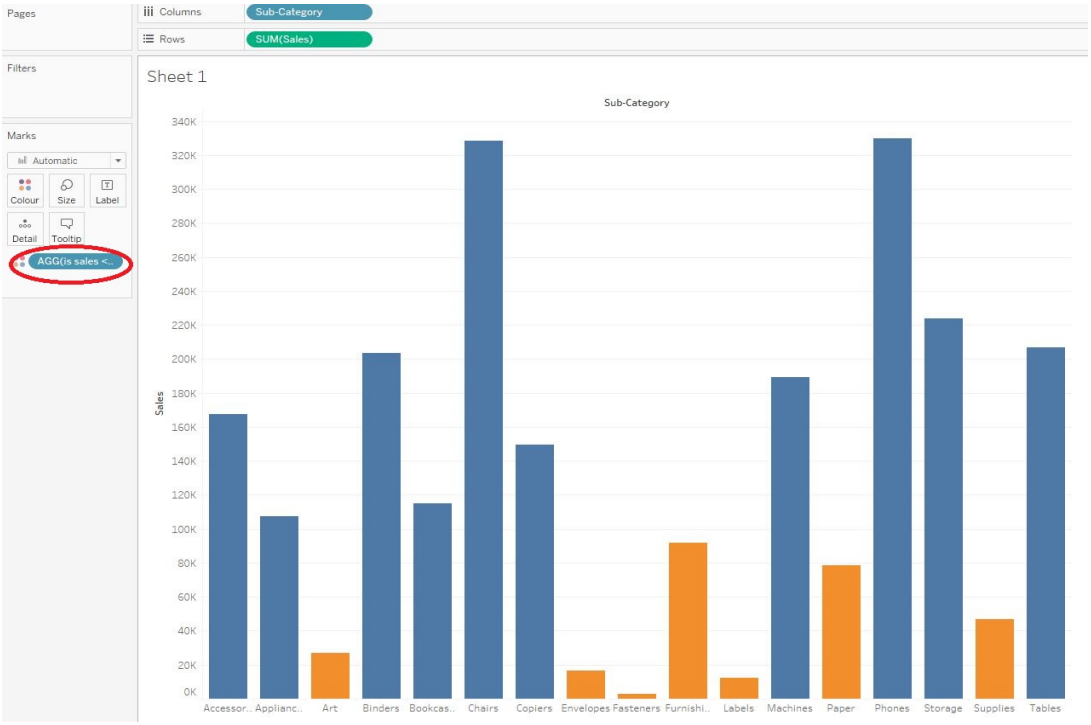


After clicking OK button, a new field of is sales < 100K? you just modified appears in the Measures area.



Step 5. Display bar chart with 2 different colors

Drag the is sales < 100K? field from the Measures pane and drop it into the Color shelf in the Marks card to color the bar chart:



Here we use two kinds of different colors to represent two sub-categories' sales: orange bars representing sub-categories having sales below 100K and blue bars for sub-categories having sales above 100K. For this purpose, we use two conditions: **IF…ELSE** function**.**

This is difference between sales and SUM (sales). Remember that SUM function returns the sum of all the values in the expression. So, SUM([Sales]) returns the total sales value for whatever dimension is being displayed in the view. Sales, on the other hand, refers to the transactional Sales.

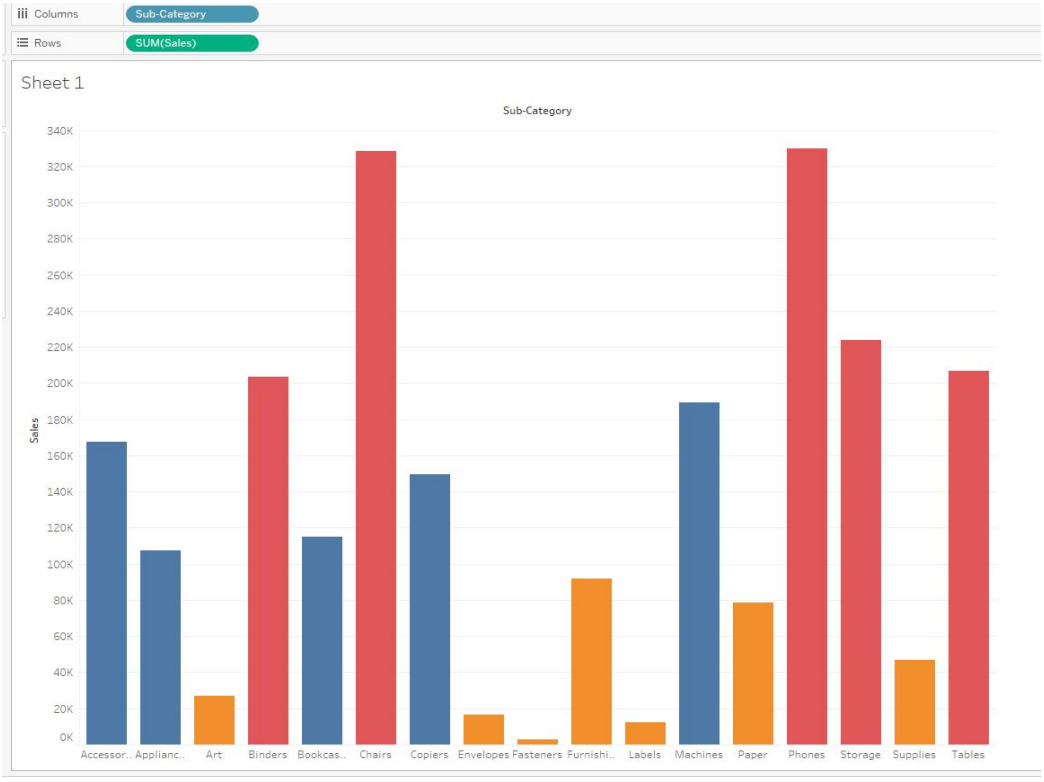
Step 6. Display bar chart with 3 different colors

If there are multiple conditions, we use IF…ELSEIF function.

Create a new calculated field called sales with 3 color:



Then drag the sales with 3 color field from the Measures pane and drop it into the Color shelf in the Marks card to color the bar chart:



Project 4 Create a Professional Report

Summarize the above experiments procedure, results, answering for questions and screenshots (project 1, 2, 3) into one report. Your report is the assignment that is required to be submitted for evaluation on week 8. Create a report by following the below steps.

Please note that

- as you may have a different version of Microsoft Word, the **following steps are just for your references.**

You can add a chapter called Chapter 6 in your previous report.

1. Open your last week's report and find the end of last week's report.
2. Copy this week's related experimental results, your findings and Screenshots, and paste them at the end of last week's report.
3. Delete original Table of Content you created.
4. Select all content , align all text to both left and right margin
5. Use shortcut key approach, generate Chapter 6: **Complex data visualizations (1)**
6. Then use shortcut key approach to generate proper sub-chapters for this week's lab work.
7. Insert Table of Contents to your report.